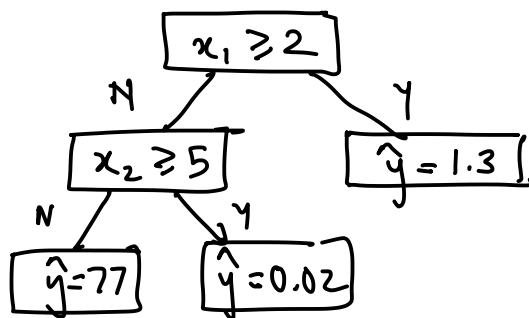


Lecture 21

Monday, April 27, 2026 7:21 PM

Recap: Decision trees for classification ✓

Today: Decision trees for regression, ensemble methodsBuildTree: Greedy & recursive algorithm
(Gini impurity index)Decision trees have high-variance
sensitive to training dataDT classification
Given x , output/predict $\hat{y} \in \{0, 1, \dots, K-1\}$ DT regression:
Given x , output $\hat{y} \in \mathbb{R}$ How to use DT for regression? Say $d=2$ 

$$x = (3, 6)$$

$$\hat{y} = 1.3$$

How to build a decision tree for regression?

SSE: Sum of squared error (TSS)

For a set I , $\bar{y} = \frac{1}{|I|} \sum_{i \in I} y^{(i)}$ ← mean of y values in I

$$SSE(I) = \sum_{i \in I} (y^{(i)} - \bar{y})^2$$

$\underbrace{\text{mean}}_{\text{Square error}}$
 $\underbrace{\text{sum of square error}}$

SSE very small $\Rightarrow y^{(i)}$ s are closer to each other
 SSE very large $\Rightarrow y^{(i)}$ are far apart from each other

$$I \rightarrow I^-, I^+$$

$$I^- \cup I^+ = I$$

$$I^- \cap I^+ = \phi$$

classification
 regression

$\downarrow GI$, good set
 $\downarrow SSE$, good set

Ensembling

- Use multiple ML models to make one "better" predictor.

→ Joy example:
 - Binary Classification
 → $\underbrace{P(\text{error}) = \epsilon}$

Case II

→ $\boxed{3 \text{ datasets}}$ 3 models
 $P(\text{error of any one model}) = \epsilon$

→ Models independent of each other
 majority voting

Case II

$$P(2 \text{ models being } x) = \binom{3}{2} \epsilon^2 (1-\epsilon)$$

$$P(3 \text{ model being } x) = \epsilon^3$$

$$P(\text{prediction } x) = 3\epsilon^2(1-\epsilon) + \epsilon^3$$

Case I $P(\text{prediction } x) = \epsilon$

$$\epsilon > 3\epsilon^2(1-\epsilon) + \epsilon^3$$

$$\Rightarrow \epsilon(1 - 3\epsilon(1-\epsilon) - \epsilon^2) > 0$$

$$\Rightarrow \epsilon(1 - 3\epsilon + 3\epsilon^2 - \epsilon^2) > 0$$

$$\Rightarrow \epsilon(1 - 3\epsilon + 2\epsilon^2) > 0$$

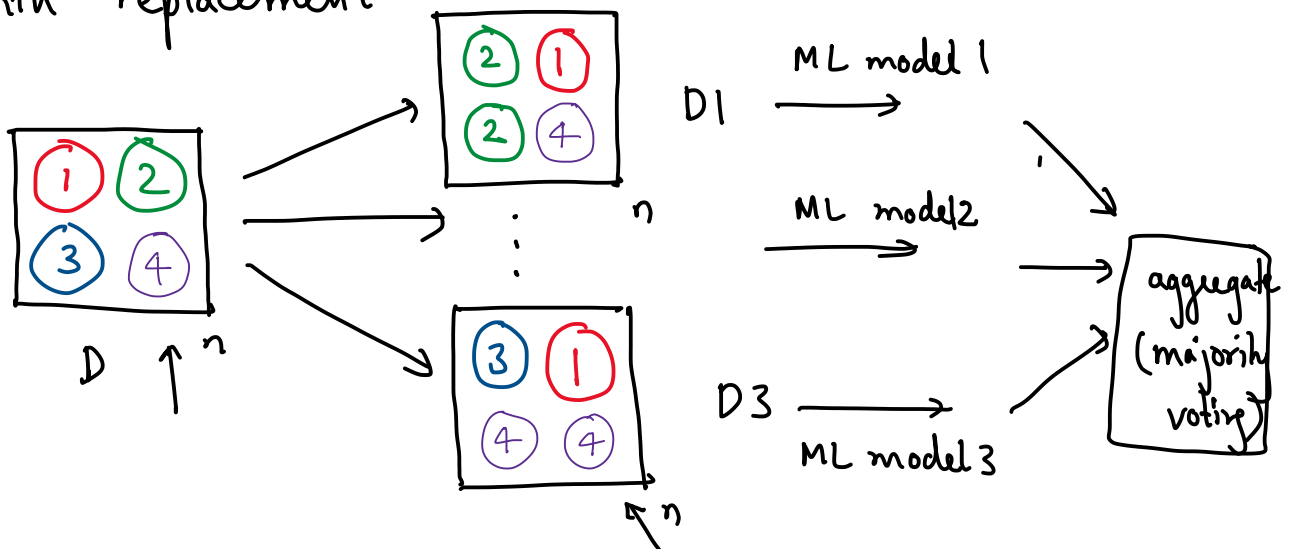
$$\Rightarrow \epsilon(2\epsilon - 1)(\epsilon - 1) > 0$$

(assume $\epsilon < 1/2$)

Case I error probability > Case II error probability

- $\{x^{(i)}, y^{(i)}\}_{i=1}^N$: data set
- We want to train B models, each on "independent data".
 - Option 1: split data into B parts
 - low data, $\uparrow$ training error
 - Simulate having multiple dataset: (Bootstrapping)

$\{x^{(i)}, y^{(i)}\}_{i=1}^N$
 - Create B dataset from this by uniformly sampling with replacement



$D1 \rightarrow$ decision tree 1
 $D2 \rightarrow$ decision tree 2 logistic regression
 $D3 \rightarrow$ decision tree 3 naive bayes

$x \rightarrow$ DT1 $\rightarrow \hat{y}_1$
 $\quad \quad \quad \rightarrow$ DT2 $\rightarrow \hat{y}_2$
 $\quad \quad \quad \rightarrow$ DT3 $\rightarrow \hat{y}_3$

$\hat{y} = \text{majority}(\hat{y}_1, \hat{y}_2, \hat{y}_3)$

Bagging (Bootstrap aggregating)

- For $b = 1, 2, \dots, B$
 - Create dataset by uniformly sampling with replacement

⇒ b) train your favourite ML model. ←

2. Prediction:

→ Classification

→ Regression

$$\hat{y} = \text{majority } (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_B)$$

$$\hat{y} = \text{average } (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_B)$$

Random Forest = Bagging + decision trees + extra randomness

1. For $b = 1, 2, \dots, B$

a) — " —

b) Decision trees

For every recursive call to Build tree algo

- - pick m out of d features, without replace
- - pick best split among only these m features

2. Prediction using aggregation.

$d = 10$ $m = 2$

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

